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Touch panel built-in display device, and method for controlling touch panel built-in display device

Abstract

A touch panel built-in display device comprises a touch coordinate detection circuit. The touch coordinate detection circuit, as to each detection electrode pair, determines presence or absence of a touch by a pointer at a coordinate in a second direction where the detection electrode pair is positioned, based on a total value or a difference value of a detection values of the detection electrode pair, and when determining that a touch is present, calculates a coordinate in the first direction of the touch by the pointer, based on a ratio of a detection value of a first detection electrode of the detection electrode pair at which a touch is determined to be present with respect to the total value of the detection electrode pair.

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Background/Summary

BACKGROUND

1. Field

(1) The present disclosure relates to a touch panel built-in display device, and a method for controlling a touch panel built-in display device.

2. Description of the Related Art

(2) JP-A-2010-113502 discloses a touch panel that includes microcapsules in which charged particles and a liquid dispersion medium are encapsulated. This touch panel includes a transparent electrode formed on a touch panel substrate, a back electrode, and a microcapsule electrophoretic display section arranged between the transparent electrode and the back electrode. The transparent electrode is connected with a voltage generation section that generates a position detection voltage and a common voltage, and a position detection section that detects a contact position based on a change in the capacitance of the transparent electrode. In the touch panel, for detecting a touch, the position detection voltage is applied to the transparent electrode, and for rewriting the display of the display section, the common voltage is applied to the transparent electrode. When a touch pen comes into contact with the touch panel substrate while the position detection voltage from the voltage generation section is applied to the transparent electrode, the capacitances at the four corners of the transparent electrode change. Based on this change in capacitance, the position detection section detects the contact position of the touch pen, and sends a predetermined signal as a detection result to a control section.

(3) The touch panel equipped with an electrophoretic display section described in JP-A-2010-113502 is capable of detecting a single contact position of the touch pen (pointer) based on the capacitances at the four corners of the transparent electrode. However, it cannot detect multiple contact positions (touch coordinates) by a pointer(s).

(4) It is an object of the present disclosure to provide a touch panel built-in display device that includes an electrophoretic display section capable of detecting a plurality of touch coordinates, and a method for controlling a touch panel built-in display device.

SUMMARY

(5) To achieve the above-described object, a touch panel built-in display device according to a first embodiment of the present disclosure includes: a plurality of pixel electrodes; a counter electrode arranged to face the plurality of pixel electrodes; an electrophoretic display section that includes charged particles arranged between the plurality of pixel electrodes and the counter electrode; a plurality of detection electrode pairs, each pair including a first detection electrode connected to the counter electrode in an edge part thereof on a first direction side, and a second detection electrode connected to the counter electrode in an edge part thereof on an opposite direction side with respect to the first direction side, the plurality of detection electrode pairs being arrayed in a second direction that crosses the first direction in plan view; and a touch coordinate detection circuit that detects a coordinate in the first direction and a coordinate in the second direction of a touch by a pointer, wherein the touch coordinate detection circuit supplies a voltage to the plurality of detection electrode pairs, acquires a detection value that is a value of current flowing through each of the detection electrodes of the plurality of detection electrode pairs, or a value of potential of the detection electrode, as to each detection electrode pair, determines presence or absence of a touch by a pointer at the coordinate in the second direction where the detection electrode pair is positioned, based on a total value or a difference value of the detection values of the detection electrode pair, and when determining that a touch is present, calculates the coordinate in the first direction of the touch by the pointer, based on a ratio of the detection value of the first detection electrode of the detection electrode pair at which a touch is determined to be present with respect to the total value of the detection electrode pair.

(6) A method for controlling a touch panel built-in display device according to a second embodiment is a method for controlling a touch panel built-in display device that includes: a plurality of pixel electrodes; a counter electrode arranged to face the plurality of pixel electrodes; an electrophoretic display section that includes charged particles arranged between the plurality of pixel electrodes and the counter electrode; a plurality of detection electrode pairs, each pair being

composed of a first detection electrode connected to the counter electrode in an edge part thereof on a first direction side, and a second detection electrode connected to the counter electrode in an edge part thereof on an opposite direction side with respect to the first direction side, the detection electrode pairs being arrayed in a second direction that crosses the first direction in plan view, the method including the steps of: supplying a voltage to the plurality of detection electrode pairs; acquiring a detection value that is a value of current flowing through each of the detection electrodes of the plurality of detection electrode pairs, or a value of potential of the detection electrode; as to each detection electrode pair, determining presence or absence of a touch by a pointer at the coordinate in the second direction where the detection electrode pair is positioned, based on a total value or a difference value of the detection values of the detection electrode pair; and when determining that a touch is present, calculating the coordinate in the first direction of the touch by the pointer, based on a ratio of the detection value of the first detection electrode of the detection electrode pair at which a touch is determined to be present with respect to the total value of the detection electrode pair.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a block diagram illustrating a configuration of a touch panel built-in display device **100** according to one embodiment.
- (2) FIG. 2 is a block diagram illustrating a configuration of a control circuit **2**.
- (3) FIG. 3 is a cross-sectional view illustrating a configuration of a touch panel **1** of the touch panel built-in display device **100**.
- (4) FIG. 4 is a perspective view schematically illustrating a configuration of the touch panel **1**.
- (5) FIG. 5 is a diagram for explaining connection relationship between a counter electrode **21** and detection electrodes **41** and **42**.
- (6) FIG. 6 is a diagram for explaining connection relationship between the counter electrode **21** and a touch coordinate detection circuit **2b**.
- (7) FIG. 7 is a diagram illustrating exemplary total values of a voltage value V1 and a voltage value V2 acquired by the touch coordinate detection circuit **2b**.
- (8) FIG. 8 is a diagram illustrating exemplary ratios of the voltage value V2 relative to the total value of the voltage values V1 and V2 acquired by the touch coordinate detection circuit **2b**.
- (9) FIG. 9 is a diagram for explaining a signal supplied from the control circuit **2** to the counter electrode **21**.
- (10) FIG. 10 is a diagram for explaining a state of connection between the counter electrode **21** and the control circuit **2**.

DESCRIPTION OF THE EMBODIMENTS

- (11) One embodiment of the present disclosure is described below in detail, with reference to the drawings. Identical or equivalent parts in the drawings are denoted by the same reference numerals, and the descriptions of the same are not repeated. To make the description easy to understand, in the drawings referred to hereinafter, the configurations are simply illustrated or schematically illustrated, or the illustration of part of constituent members is omitted. Further, dimension ratios of constituent members illustrated in the drawings do not necessarily indicate actual dimension ratios.
- (12) [Overall Configuration of Touch Panel Built-In Display Device]
- (13) A configuration of a touch panel built-in display device **100** according to the present embodiment (hereinafter referred to as “display device **100**”) is described. FIG. 1 is a block diagram illustrating a configuration of the display device **100** according to the present embodiment. FIG. 2 is a block diagram illustrating a configuration of the control circuit **2**. FIG. 3 is a cross-sectional view illustrating a configuration of a touch panel **1** of the display device **100**.

(14) As illustrated in FIG. 1, the display device **100** includes the touch panel **1** and the control circuit **2**. In the present embodiment, the display device **100** is configured as an electronic paper, an electrophoretic display, and a non-emissive display. The touch panel **1** displays images or videos based on control signals from the control circuit **2**. Furthermore, in the present embodiment, the touch panel **1** detects multiple touched positions (touch coordinates) touched by a pointer(s). The control circuit **2** outputs a control signal to the touch panel **1** and controls the operation of the touch panel **1**.

(15) As illustrated in FIG. 2, the control circuit **2** includes a device control circuit **2a**, a touch coordinate detection circuit **2b**, and a timing control circuit **2c**. Each of the device control circuit **2a**, the touch coordinate detection circuit **2b**, and the timing control circuit **2c** is formed as, for example, a System-On-Glass (SOG) formed on a pixel substrate **10** (see FIG. 3). The device control circuit **2a** performs a controlling operation of rewriting an image displayed on the touch panel **1**. The touch coordinate detection circuit **2b** outputs a voltage to the touch panel **1**, acquires detection values from the touch panel **1**, and detects a touch(es) (contact(s)) on the touch panel **1** by a pointer(s) (finger(s) or pen(s)) based on the detection value. The touch coordinate detection circuit **2b** transmits a touch detection result (report) to the control circuit **2**. The device control circuit **2a** obtains a report of touch coordinates of the position(s) touched by the pointer(s), from the touch coordinate detection circuit **2b**, and controls the display device **100** according to the report. The timing control circuit **2c** switches between a state in which the touch panel **1** (counter electrode **21**) and the device control circuit **2a** are connected and a state in which the touch panel **1** (counter electrode **21**) and the touch coordinate detection circuit **2b** are connected.

(16) As illustrated in FIG. 3, the touch panel **1** includes a pixel substrate **10**, a counter substrate **20**, an electrophoretic display section **30**, detection electrodes **41**, and detection electrodes **42**. The pixel substrate **10** and the counter substrate **20** are flexible substrates. In addition, the pixel substrate **10** is a transparent substrate that transmits visible light. The pixel substrate **10** and the counter substrate **20** are, for example, resin film substrates of polyethylene terephthalate (PET). A plurality of pixel electrodes **11** and a plurality of lines **12** are arranged on the pixel substrate **10**. The pixel electrodes **11** are made of a metal or a conductor such as indium tin oxide (ITO). The pixel electrodes **11** are arranged in matrix in plan view. Further, the lines **12** are formed of a metal (copper, gold, silver, etc.) on a surface of the pixel substrate **10**, for example.

(17) The counter substrate **20** is arranged to face the pixel substrate **10** in a normal direction with respect to the pixel electrode **10** (in Z1 direction in FIG. 2). A counter electrode **21** is arranged on the counter substrate **20**. The counter electrode **21** is arranged to face the pixel electrodes **11**. That is, the counter electrode **21** is one common electrode provided in common to the pixel electrodes **11**. The counter electrode **21** is a transparent electrode that transmits visible light. As the counter electrode **21**, for example, indium tin oxide (ITO) can be used. The counter electrode **21** is connected to the lines **12** via the detection electrodes **41** or the detection electrodes **42**.

(18) The electrophoretic display section **30** is arranged between the pixel substrate **10** and the counter substrate **20**. The electrophoretic display section **30** includes a plurality of microcapsules **31**, a plurality of charged particles **32a** and **32b**, a dispersion medium **33**, and an insulating layer **34**. The microcapsules **31** are arranged within the insulating layer **34**. The charged particles **32a** and **32b** and the dispersion medium **33** are enclosed within the microcapsules **31**. The charged particles **32a** and **32b** include white charged particles **32a** and black charged particles **32b**. For example, the white charged particles **32a** are positively charged, and the black charged particles **32b** are negatively charged. The dispersion medium **33** is liquid with insulating properties such as, for example, an organic solvent. The insulating layer **34** fills a space between the pixel substrate **10** and the counter substrate **20**. The insulating layer **34** is, for example, a layer made of a polymeric resin material.

(19) Then, by applying a predetermined voltage across the pixel electrodes **11** and the counter electrode **21**, when the pixel electrodes **11** become positive electrodes and the counter electrode **21**

becomes a negative electrode, the black charged particles **32b** are attracted to the pixel electrodes **11**, and the white charged particles **32a** are attracted to the counter electrode **21**. In this case, white color is displayed by the touch panel **1** to the user observing from the counter substrate **20** side. Furthermore, as illustrated in FIG. 3, when the pixel electrodes **11** become negative electrodes and the counter electrode **21** becomes a positive electrode, the black charged particles **32b** are attracted to the counter electrode **21**, and the white charged particles **32a** are attracted to the pixel electrode **11**. In this case, black color is displayed by the touch panel **1** to the user observing from the counter substrate **20** side.

(20) The detection electrodes **41** and the detection electrodes **42** are, for example, common transfer electrodes arranged between the pixel substrate **10** and the counter substrate **20**. The detection electrodes **41** and the detection electrodes **42** include conductive particles arranged within a resin material, and thereby have conductivity. With this configuration, the detection electrodes **41** and the detection electrodes **42** electrically connect the counter electrode **21** and the lines **12**.

(21) FIG. 4 is a perspective view schematically illustrating a configuration of the touch panel **1**. Here, the X axis is an axis on a plane of the touch panel **1**. The Y axis is an axis on the plane of the touch panel **1**, and is perpendicular to the X axis. An X1 direction is one direction parallel to the X axis, and an X2 direction is a direction opposite to the X1 direction. A Y1 direction is one direction parallel to the Y axis, and a Y2 direction is a direction opposite to the Y1 direction. As illustrated in FIG. 4, the detection electrodes **41** are connected to the counter electrode **21** in an edge part **21a** thereof on the X1 direction side. The “edge part” means not only an end portion of the electrode, but also an area within a predetermined range from the end portion. The area within the predetermined range is, for example, an area of the counter electrode **21** that is outside the electrophoretic display section **30** in plan view.

(22) The detection electrodes **41** are arranged at intervals in a direction parallel to the Y axis. For example, the detection electrodes **41** are arranged at equal intervals in the Y1 direction. Further, the detection electrodes **42** are connected to the counter electrode **21** in an edge part **21b** thereof on the X2 direction side. The detection electrodes **42** are arranged at intervals in the Y1 direction. For example, the detection electrodes **42** are arranged at equal intervals in the Y1 direction. The number of the detection electrodes **41** is the same as the number of the detection electrodes **42**. Regarding resistances illustrated in FIG. 4, they are schematic illustrations of internal resistances in the counter electrode **21**, and do not indicate that resistors are arranged on the counter electrode **21**.

(23) FIG. 5 is a diagram for explaining connection relationship between the counter electrode **21** and the detection electrodes **41** and **42**. As illustrated in FIG. 5, to distinguish the detection electrodes **41** from one another, the detection electrodes **41** are referred to as detection electrodes **41a** to **41h** in the order directed in the Y2 direction. Further, to distinguish the detection electrodes **42** from one another, the detection electrodes **42** are referred to as detection electrodes **42a** to **42h** in the order directed in the Y2 direction. Here, the detection electrode **41a** has the same coordinate in the direction parallel to the Y axis (referred to as the “Y coordinate”) as that of the detection electrode **42a**. The detection electrodes **41b** to **41h** have the same Y coordinates as those of the detection electrodes **42b** to **42h**, respectively. Here, two detection electrodes having the same Y coordinate (for example, the detection electrode **41a** and the detection electrode **42a**) are referred to as a “detection electrode pair.” The detection electrodes **41b** to **41h** form detection electrode pairs with the detection electrodes **42b** to **42h**, respectively.

(24) FIG. 6 is a diagram for explaining connection relationship between the counter electrode **21** and the touch coordinate detection circuit **2b**. As illustrated in FIG. 6, a resistance value of the counter electrode **21** between the detection electrode **41** and the detection electrode **42** is given as R. In other words, any of the resistance values of the counter electrode **21** between the detection electrodes **41a** to **41h** and the detection electrodes **42a** to **42h** is R. In addition, any of respective resistance values between the detection electrodes **41a** to **41h** and the detection electrodes **42a** to **42h** is R_c. R_c is smaller than R.

(25) As illustrated in FIG. 3, the detection electrodes **41** and **42** are connected to a power supply circuit **51**, as well as two current detectors **52**, respectively, via the lines **12**. Specifically, as illustrated in FIG. 6, the current detector **52** includes a plurality of shunt resistors **52a**, selection circuits **52ba** and **52bb**, and voltmeters **52ca** and **52cb**. The shunt resistors **52a** are arranged between the power supply circuit **51** and the detection electrodes **41a** to **41h**, respectively. Furthermore, the shunt resistors **52a** are arranged between the power supply circuit **52** and the detection electrodes **42a** to **42h**, respectively. The resistance value of the shunt resistor **52a** is R_s . R_s is smaller than R .

(26) The power supply circuit **51** is connected to the detection electrodes **41a** to **41h** and the detection electrodes **42a** to **42h** via one of the shunt resistors **52a**. Then, the power supply circuit **51** outputs a driving voltage. For example, the driving voltage is a sine wave (alternating voltage) having a predetermined amplitude and a predetermined frequency. In this configuration, no current flows through the counter electrode **21** when the pointer is not touching the counter electrode **21**. However, when the pointer touches the counter electrode **21**, the touched point becomes grounded, and thereby a current flows from the detection electrode pair (detection electrodes **41** and **42**) toward the touched point. Therefore, the display device **100** according to the present embodiment identifies the touched point (coordinates) by detecting the voltage value (detection value) corresponding to the above-mentioned current.

(27) The selection circuit **52ba** selects the connection point connected to the voltmeter **52ca**, from among the respective connection points connected between the detection electrodes **41a** to **41h** and the shunt resistor **52a**. For example, the selection circuit **52ba** sequentially switches the connection point (scans the connection points) connected to the voltmeter **52ca** according to the control signal supplied from the touch coordinate detection circuit **2b**. Then, the voltmeter **52ca** detects the voltage value and transmits the detection result to the touch coordinate detection circuit **2b**.

(28) The selection circuit **52bb** selects the connection point connected to the voltmeter **52cb** from among the respective connection points connected between the detection electrodes **42a** to **42h** and the shunt resistor **52a**. For example, the selection circuit **52bb** sequentially switches (scans) the connection points connected to the voltmeter **52cb** according to the control signal supplied from the touch coordinate detection circuit **2b**. The selection circuit **52bb** selects a connection point connected to the detection electrode **42** having the same Y coordinate as the detection electrode **41** connected to the connection point selected by the selection circuit **52ba**.

(29) Here, in the present embodiment, the touch coordinate detection circuit **2b** acquires voltage values from the voltmeters **52ca** and **52cb** while the driving voltage is supplied from the power supply circuit **51** to the detection electrodes **41a** to **41h** and the detection electrodes **42a** to **42h**. Here, the voltage value detected by the voltmeter **52ca** is given as $V1$. The voltage value detected by the voltmeter **52cb** is given as $V2$. A connection point detected by the selection circuit **52ba** and the selection circuit **52bb** at the n -th time (n is a natural number) is given as CL_n (clock n). Then, the touch coordinate detection circuit **2b** acquires a difference value ($V2-V1$) between the voltage value $V1$ and the voltage value $V2$ from the voltmeters **52ca** and **52cb**. When the difference value ($V2-V1$) is not 0, the touch coordinate detection circuit **2b** assumes that there has been a touch, and the touch coordinate detection circuit **2b** acquires a total value ($V1+V2$) of the voltage value $V1$ and the voltage value $V2$. It should be noted that when the difference value ($V2-V1$) is 0, the touch coordinate detection circuit **2b** determines that there is no touch.

(30) FIG. 7 is a diagram illustrating exemplary total values of a voltage value $V1$ and a voltage value $V2$ acquired by the touch coordinate detection circuit **2b**. The example in FIG. 7 shows total values of the voltage value $V1$ and the voltage value $V2$ at $CL1$ to $CL8$. The touch coordinate detection circuit **2b** detects CL that is the peak value, from the total values of the voltage value $V1$ and the voltage value $V2$ at $CL1$ to $CL8$. The touch coordinate detection circuit **2b** detects, for example, a $CL(s)$ at which a larger value than that at an adjacent CL ($CL2$ and $CL6$ in the example of FIG. 7) is detected, as a $CL(s)$ at which a peak value is detected. As a result, the touch coordinate

detection circuit **2b** determines that there are touch(es) by pointer(s) at positions of the Y coordinates of the detection electrodes **41b** and **42b** corresponding to CL2. Then, the touch coordinate detection circuit **2b** determines that the positions of the Y coordinates of the detection electrodes **41b** and **42b** corresponding to CL2 are the Y coordinates of the touch(es) by the pointer(s). For example, the touch coordinate detection circuit **2b** assumes values obtained by multiplying the number given to the clock ("2" in the case of CL2) by predetermined values (the widths of the detection electrodes **41** and **42** in the direction parallel to the Y axis), as the Y coordinates (Yp).

(31) FIG. **8** is a diagram illustrating exemplary ratios of the voltage value V2 relative to the total value of the voltage values V1 and V2 acquired by the touch coordinate detection circuit **2b**. The example in FIG. **8** shows a ratio of the voltage value V2 with respect to the total value of the voltage value V1 and the voltage value V2 at each of CL1 to CL8. Then, the touch coordinate detection circuit **2b** calculates the ratio of the voltage value V2 with respect to the total value of the voltage value V1 and the voltage value V2, $P_x (=V2/(V1+V2))$, in each of CL1 to CL8.

(32) The touch coordinate detection circuit **2b** determines X coordinate(s) (Xp) of a touch(es) by the pointer(s), using the ratio P_x of the detection electrodes **41** and **42** at which it is determined that there are touches, based on the following formula (1):

$$(33) \quad X_p = (P_x - C_0) / K_0 \times W_x \quad (1)$$

(34) where W_x is the width in the direction parallel to the X axis of the touch panel **1**.

(35) Here, C_0 and K_0 are values obtained by taking the influences of the resistance (line resistance) of the detection electrode **41** or **42** and the resistance of the shunt resistor **52a** into consideration, which are the values shown in the following equations (2) and (3):

$$(36) \quad C_0 = (R_s + R_c) / (2R_s + 2R_c + R) \quad (2) \quad K_0 = R / (2R_s + 2R_c + R) \quad (3)$$

(37) With this configuration, even if the resistance (line resistance) of the detection electrode **41** or **42** and the resistance of the shunt resistor **52a** are large enough to not be ignored, the display device **100** can calculate accurate X coordinates (Xp).

(38) The touch coordinate detection circuit **2b** transmits information (report) on the calculated X and Y coordinates (in the example of FIG. **6**, the touch coordinates of two points (P1 and P2)) to the device control circuit **2a**.

(39) With the present embodiment, the presence or absence of a touch can be determined for each of the detection electrodes **41** and **42** arranged at different positions in the direction parallel to the Y axis. Therefore, even if the touch panel **1** is touched by pointer(s) at a plurality of positions that are different in the direction parallel to the Y axis, each of touches at a plurality of positions can be detected. Then, when it is determined that there are touches, the X coordinates of each of the multiple touched positions can be calculated, whereby touch coordinates at the multiple touched positions can be detected.

(40) FIG. **9** is a diagram for explaining a signal supplied from the control circuit **2** to the counter electrode **21**. In FIG. **9**, the potential supplied to the counter electrode **21** is shown as V_c . As illustrated in FIG. **9**, during the touch detection period T1, the control circuit **2** applies a voltage from the power supply circuit **51** to the counter electrode **21**, and the touch coordinate detection circuit **2b** detects the touch coordinates. During the display period T2, the control circuit **2** applies a common voltage V_c , that is, alternately applies a positive voltage and a negative voltage to the counter electrode **21**. Then, image signals are supplied by the control circuit **2** to the pixel electrodes **11**, and the image displayed is rewritten by the electrophoretic display section **30**. The timing control circuit **2c** of the control circuit **2** switches the touch detection period T1 and the display period T2 at predetermined time intervals. As a result, touch detection processing and display processing are executed in a time-division manner. This makes it possible to use the counter electrode **21** for both touch detection and display, thereby making it unnecessary to provide an electrode separate from (in a different layer from) the counter electrode **21**.

(41) FIG. 10 is a diagram for explaining a state of connection between the counter electrode 21 and the control circuit 2. As illustrated in FIG. 10, during the touch detection period T1, the counter electrode 21 is connected to the touch coordinate detection circuit 2b. During the display period T2, the counter electrode 21 is connected to the device control circuit 2a. It should be noted that at time t2 when the touch detection period T1 switches to the display period T2, a report (touch coordinates) is transmitted from the touch coordinate detection circuit 2b to the device control circuit 2a.

Modification Example

(42) The above-described embodiment is merely an example for implementing the present disclosure. The present disclosure, therefore, is not limited to the above-described embodiment, and the above-described embodiment can be appropriately varied and implemented without departing from the spirit and scope of the disclosure.

(43) (1) In the above embodiment, an example is shown in which microcapsules are provided in the electrophoretic display section, but the present disclosure is not limited thereto. That is, the electrophoretic display section may be composed of charged particles and a dispersion medium, while microcapsules are not provided in the electrophoretic display section.

(44) (2) In the above embodiment, an example is shown in which the control circuit (device control circuit, touch coordinate detection circuit, and timing control circuit) is configured as an SOG formed on the pixel substrate, but the present disclosure is not limited to this. For example, at least a portion of the device control circuit, the touch coordinate detection circuit, and the timing control circuit may be arranged on a substrate separate from the pixel substrate.

(45) (3) In the above embodiment, an example is shown in which the potential of the detection electrode detected by the voltmeter is used as the detection value, but the present disclosure is not limited to this. That is, an ammeter that detects current flowing through the detection electrode may be arranged, based on the value of the current, the presence or absence of a touch may be determined based on the value of the current, and the X coordinate of the touch may be determined. As illustrated in FIG. 6, since the current flows toward the touched position, the total value of the current flowing through the detection electrode pair corresponding to the Y coordinate of the touch increases. With this configuration, when the display device detects the peak value of the current, the Y coordinate corresponding to the peak value can be determined as the touch coordinate, as is the case with the example using the voltage values shown in FIG. 7, and the X coordinate can be determined as the touch coordinate based on the ratio, as is the case with the example using the voltage values shown in FIG. 8.

(46) (4) In the above embodiment, an example is shown in which the selection circuit is provided, but the present disclosure is not limited to this. That is, a voltmeter may be disposed with respect to each of the detection electrodes.

(47) (5) In the above embodiment, an example is shown in which the touch coordinate detection circuit is configured to detect a CL at which a larger value than that at an adjacent CL is detected, as a CL at which a peak value is detected, but the present disclosure is not limited to this. For example, the touch coordinate detection circuit may be configured to determine that there is a touch when the total value ($V1+V2$) is equal to or higher than a threshold voltage V_{th} .

(48) (6) In the above embodiment, an example is shown in which the pixel substrate and the counter substrate are made of PET, but the present disclosure is not limited to this. For example, the pixel substrate and the counter substrate may be made of polycarbonate, acrylic, or quartz glass.

(49) (7) In the above embodiment, an example is shown in which the presence or absence of a touch(es) by a pointer(s) is determined based on whether the difference value ($V2-V1$) between the voltage values of the detection electrode pair is 0, and when the difference value ($V2-V1$) is not 0, the peak value of the total voltage value ($V2+V1$) of the detection electrode pair is used to determine that there is a touch, but the present disclosure is not limited to this. In other words, the presence or absence of a touch(s) by a pointer(s) may be determined based on only either one of the

following: whether the difference value ($V2-V1$) between voltage values of a detection electrode pair is 0 or not; and whether there is a peak of the total value ($V2+V1$) of voltage values of a detection electrode pair.

(50) The above-described configuration can also be described as follows.

(51) A touch panel built-in display device according to a first configuration includes: a plurality of pixel electrodes; a counter electrode arranged to face the plurality of pixel electrodes; an electrophoretic display section that includes charged particles arranged between the plurality of pixel electrodes and the counter electrode; a plurality of detection electrode pairs, each pair including a first detection electrode connected to the counter electrode in an edge part thereof on a first direction side, and a second detection electrode connected to the counter electrode in an edge part thereof on an opposite direction side with respect to the first direction side, the plurality of detection electrode pairs being arrayed in a second direction that crosses the first direction in plan view; and a touch coordinate detection circuit that detects a coordinate in the first direction and a coordinate in the second direction of a touch by a pointer, wherein the touch coordinate detection circuit supplies a voltage to the plurality of detection electrode pairs; acquires a detection value that is a value of current flowing through each of the detection electrodes of the plurality of detection electrode pairs, or a value of potential of the detection electrode; as to each detection electrode pair, determines presence or absence of a touch by a pointer at the coordinate in the second direction where the detection electrode pair is positioned, based on a total value or a difference value of the detection values of the detection electrode pair; and when determining that a touch is present, calculates the coordinate in the first direction of the touch by the pointer, based on a ratio of the detection value of the first detection electrode of the detection electrode pair at which a touch is determined to be present with respect to the total value of the detection electrode pair (the first configuration).

(52) With the above-described first configuration, the presence or absence of a touch(es) can be determined for each of the detection electrode pairs arranged at different positions in the second direction, and therefore, even if the touch panel is touched by a pointer(s) at a plurality of positions that are different in the second direction, each of touches at a plurality of positions can be detected. Then, when it is determined that there is a touch(es), the X coordinates in the first direction of each of the multiple touched positions can be calculated, whereby touch coordinates at the multiple touched positions can be detected.

(53) In the first configuration, the touch coordinate detection circuit may include: a resistor connected to the first detection electrode; and a voltmeter connected to the resistor, and the touch coordinate detection circuit may acquire a voltage value detected by the voltmeter as a detection value of the first detection electrode (the second configuration).

(54) With the above-described second configuration, the detection value of the first detection electrode can be acquired by detecting a voltage value.

(55) In the second configuration, the touch coordinate detection circuit may set $C0$ as a value given by the following formula (1a), and set $K0$ as a value given by the following formula (2a), wherein a coordinate X in the first direction of the touch by the pointer is calculated based on the following formula (3a) (the third configuration):

$$(56) \quad C0 = (Rs + Rc) / (2Rs + 2Rc + R) \quad (1a) \quad K0 = R / (2Rs + 2Rc + R) \quad (2a)$$

(57) where R represents a resistance value of the counter electrode, Rc represents a resistance value of the detection electrode, and Rs represents a resistance value of the resistor,

$$(58) \quad X = (Px - C0) / K0 \times Wx \quad (3a)$$

(59) where Px represents a ratio of the detection value of the first detection electrode of the detection electrode pair with respect to the total value, and Wx represents a predetermined constant.

(60) With the above-described third configuration, even when the resistance value of the detection electrode and the resistance value of the resistor are large enough to not be ignored, the coordinate

in the first direction can be calculated accurately.

(61) In the first configuration, the touch coordinate detection circuit may set a value obtained by multiplying a ratio of the detection value of the first detection electrode of the detection electrode pair with respect to the total value, by a predetermined constant, as a coordinate in the first direction of the touch by the pointer (the fourth configuration).

(62) With the above-described fourth configuration, the calculation for determining the coordinate in the first direction can be simplified.

(63) In any one of the first to fourth configurations, the touch panel may further include: a display drive circuit that supplies a display signal to the counter electrode; and a timing control circuit that allows supply of the display signal executed by the display drive circuit, and detection of a coordinate in the first direction and a coordinate in the second direction of a touch by a pointer executed by the touch coordinate detection circuit, to be executed in a time-division manner (the fifth configuration).

(64) With the above-described fifth configuration, in which the period for displaying an image on the touch panel and the period for detecting a touch can be divided, it is possible to prevent the display of an image from affecting the detection of a touch, and also to prevent the detection of a touch from affecting the display of an image.

(65) A method for controlling a touch panel built-in display device according to a sixth configuration is a method for controlling a touch panel built-in display device that includes: a plurality of pixel electrodes; a counter electrode arranged to face the plurality of pixel electrodes; an electrophoretic display section that includes charged particles arranged between the plurality of pixel electrodes and the counter electrode; a plurality of detection electrode pairs, each pair including a first detection electrode connected to the counter electrode in an edge part thereof on a first direction side, and a second detection electrode connected to the counter electrode in an edge part thereof on an opposite direction side with respect to the first direction side, the plurality of detection electrode pairs being arrayed in a second direction that crosses the first direction in plan view, the method including the steps of: supplies a voltage to the plurality of detection electrode pairs; acquiring a detection value that is a value of current flowing through each of the detection electrodes of the plurality of detection electrode pairs, or a value of potential of the detection electrode; as to each detection electrode pair, determining presence or absence of a touch by a pointer at the coordinate in the second direction where the detection electrode pair is positioned, based on a total value or a difference value of the detection values of the detection electrode pair; and when determining that a touch is present, calculating the coordinate in the first direction of the touch by the pointer, based on a ratio of the detection value of the first detection electrode of the detection electrode pair at which a touch is determined to be present with respect to the total value of the detection electrode pair (the seventh configuration).

(66) With the above-described sixth configuration, it is possible to provide a method for controlling a touch panel built-in display device that includes an electrophoretic display section capable of detecting touch coordinates at the multiple touched positions.

(67) The present disclosure contains subject matter related to that disclosed in Japanese Priority Patent Application JP 2023-070049 filed in the Japan Patent Office on Apr. 21, 2023, the entire contents of which are hereby incorporated by reference.

(68) It should be understood by those skilled in the art that various modifications, combinations, sub-combinations and alterations may occur depending on design requirements and other factors insofar as they are within the scope of the appended claims or the equivalents thereof.

Claims

1. A touch panel built-in display device comprising: a plurality of pixel electrodes; a counter electrode arranged to face the plurality of pixel electrodes; an electrophoretic display section that

includes charged particles arranged between the plurality of pixel electrodes and the counter electrode; a plurality of detection electrode pairs, each pair being composed of a first detection electrode connected to the counter electrode in an edge part thereof on a first direction side, and a second detection electrode connected to the counter electrode in an edge part thereof on an opposite direction side with respect to the first direction side, the detection electrode pairs being arrayed in a second direction that crosses the first direction in plan view; and a touch coordinate detection circuit that detects a coordinate in the first direction and a coordinate in the second direction of a touch by a pointer, wherein the touch coordinate detection circuit supplies a voltage to the plurality of detection electrode pairs; acquires a detection value that is a value of current flowing through each of the detection electrodes of the plurality of detection electrode pairs, or a value of potential of the detection electrode; as to each detection electrode pair, determines presence or absence of a touch by a pointer at the coordinate in the second direction where the detection electrode pair is positioned, based on a total value or a difference value of the detection values of the detection electrode pair; and when determining that a touch is present, calculates the coordinate in the first direction of the touch by the pointer, based on a ratio of a detection value of the first detection electrode of the detection electrode pair at which a touch is determined to be present with respect to the total value of the detection electrode pair, the touch coordinate detection circuit includes a resistor connected to the first detection electrode, and a voltmeter connected to the resistor, the touch coordinate detection circuit acquires a voltage value detected by the voltmeter as a detection value of the first detection electrode, and the touch coordinate detection circuit sets C0 as a value given by the following formula (1), and sets K0 as a value given by the following formula (2), wherein a coordinate X in the first direction of the touch by the pointer is calculated based on the following formula (3): $C0 = (Rs + Rc) / (2Rs + 2Rc + R)$ (1) $K0 = R / (2Rs + 2Rc + R)$ (2) where R represents a resistance value of the counter electrode, Rc represents a resistance value of the detection electrode, and Rs represents a resistance value of the resistor,

$X = (Px - C0) / K0 \times Wx$ (3) where Px represents a ratio of the detection value of the first detection electrode of the detection electrode pair with respect to the total value, and Wx represents a predetermined constant.

2. The touch panel built-in display device according to claim 1, wherein the touch coordinate detection circuit sets a value obtained by multiplying a ratio of the detection value of the first detection electrode of the detection electrode pair with respect to the total value, by a predetermined constant, as the coordinate in the first direction of the touch by the pointer.

3. The touch panel built-in display device according to claim 1, the touch panel built-in display device further comprising: a display drive circuit that supplies a display signal to the counter electrode; and a timing control circuit that allows supply of the display signal executed by the display drive circuit, and detection of a coordinate in the first direction and a coordinate in the second direction of a touch by a pointer executed by the touch coordinate detection circuit, to be executed in a time-division manner.

4. A method for controlling a touch panel built-in display device, wherein the touch panel built-in display device includes: a plurality of pixel electrodes; a counter electrode arranged to face the plurality of pixel electrodes; an electrophoretic display section that includes charged particles arranged between the plurality of pixel electrodes and the counter electrode; and a plurality of detection electrode pairs, each pair being composed of a first detection electrode connected to the counter electrode in an edge part thereof on a first direction side, and a second detection electrode connected to the counter electrode in an edge part thereof on an opposite direction side with respect to the first direction side, the detection electrode pairs being arrayed in a second direction that crosses the first direction in plan view; and a resistor connected to the first detection electrode, and a voltmeter connected to the resistor, the method comprising the steps of: supplying a voltage to the plurality of detection electrode pairs; acquiring a detection value that is a value of current flowing

through each of the detection electrodes of the plurality of detection electrode pairs, or a value of potential of the detection electrode; as to each detection electrode pair, determining presence or absence of a touch by a pointer at the coordinate in the second direction where the detection electrode pair is positioned, based on a total value or a difference value of the detection values of the detection electrode pair; when determining that a touch is present, calculating the coordinate in the first direction of the touch by the pointer, based on a ratio of a detection value of the first detection electrode of the detection electrode pair at which a touch is determined to be present with respect to the total value of the detection electrode pair; and acquiring a voltage value detected by a voltmeter connected to a resistor that is connected to the first detection electrode as a detection value of the first detection electrode; wherein in determining that the touch is present, setting C0 as a value given by the following formula (1), and setting K0 as a value given by the following formula (2), wherein a coordinate X in the first direction value is calculated based on the following formula (3):

$$C0=(Rs+Rc)/(2Rs+2Rc+R) \quad (1)$$

$K0=R/(2Rs+2Rc+R)$ (2) where R represents a resistance value of the counter electrode, Rc represents a resistance value of the detection electrode, and Rs represents a resistance value of the resistor,

$X=(Px-C0)/K0 \times Wx$ (3) where Px represents a ratio of the detection value of the first detection electrode of the detection electrode pair with respect to the total value, and Wx represents a predetermined constant.
